# CSU Hackathon Geospatial Primer

## What is GIS data

- Gentle Introduction to GIS, a great resource from QGIS
- https://carpentries-incubator.github.io/geospatial-python/

## Geospatial Python Resources

https://autogis-site.readthedocs.io/en/latest/index.html

https://carpentries-incubator.github.io/geospatial-python/ (Same link as in the What is GIS data section)

## Vector

- Gentle introduction to: Vector data

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|---|---|
| Part 1: Vector and Raster overview | 13:11 |
| Part 2: Vector data structure and Properties | 10:36 |
| Part 3: Vector file formats explained | 13:40 |

## Raster

- Gentle introduction to: Raster data

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|---|---|
| Part 4: Raster data structure and properties | 23:53 |
| Part 5: Raster file formats explained | 11:57 |
| Part 6: Comparing raster and vector and their uses | 6:27 |

## Attributes and Tables

- Gentle introduction to: Attribute data (tables)

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|---|---|
| Part 6: Understanding tables | 14:50 |
| Part 7: Getting tabular data on a map | 18:07 |

## GIS Software and Platforms

Geospatial data can be worked with and analyzed in one of dozens of different platforms and languages.

This table shows some of the most common options, but there are more!

| Platform | Type | Cost | Description |
|---|---|---|---|
| **ArcGIS Pro** | Desktop | Licensed | Desktop GIS by Esri, Windows only. Available to all in CSU |
| **ArcGIS Online** | Browser-based | Licensed + Public | Web GIS by Esri, data creation, sharing, and analysis |
| **QGIS** | Desktop | Free, open source | Powerful and free, works on a Mac |
| **R** | Programming | Free, open source | Data science and statistical programming language with many spatial and mapping libraries |
| **Python** | Programming | Free, open source | Data science and automation language with many spatial and mapping libraries |
| **Tableau** | Browser-based | Subscription | Tabular data analysis with spatial options |
| **Google Earth Engine** | Browser-based | Free / Subscription | JS-based coding to access remote sensing data and processes |
| **Google Earth** | Desktop | Free | Simple mapping, data creation |
| **PostGIS** | Desktop/ Database | Free, open source | PostGres-based database extension to enable fast and powerful spatial analyses |

## How do the layers line up correctly?

Videos:

- Introducing Coordinates
- Introducing Map Projections
- Coordinate Reference Systems

**Coordinate Values for CSU Morgan Library in some common coordinate systems:**

| Coordinate System | Type | Map Units | Description | Coordinates of Morgan Library |
|---|---|---|---|---|
| **Geographic (GCS)** | Geographic/ Unprojected | Decimal Degrees | Lat-Longs. Worldwide, common, not good for area and distance calculations | -105.084501, 40.572987 |
| **UTM** | Projected | Meters | 60 UTM zones cover the world, coordinates are applied within each zone with minimal distortion | 492847.681715, 4491359.78254 |
| **State Plane** | Projected | Feet | Each US state is divided into zones, where each zone has an optimal projection applied to minimize distortion | 3115425.1024, 1451850.80408 |
| **Lambert Conformal Conic** | Projected | Meters | Spans North America, does have some distortion since it's such a large region | -721256.933338, 97370.212465 |

- When writing code, and loading layers, **be aware of the coordinate reference system**, or CRS, of each layer. If the CRSs don't match, the layers won't align, and analysis or overlay is not possible.
- In geopandas, use `.crs` to retrieve the CRS information (EPSG code) and `.to_crs` to switch to a different CRS
- In rasterio, use `.crs` to retrieve the CRS and a more complex `.transform` operation to change the CRS
- In rioxarray, use `.crs` and `.reproject`.
- Other libraries exist for this as well.

In addition to CRSs needing to match for rasters to overlay, the **extents** (number of rows and columns) also need to match. Use the rasterio **Mask** method (or equivalents in other libraries) for this.

Even if the coordinate systems do match, and the data do align, then you still need to be concerned with WHAT coordinate system it is.

- Any data in a **Geographic** Coordinate System will have large amounts of **distortion**, and calculations of area or distance will be incorrect, and plots/maps will look misshapen.
- **Projected** Coordinate Systems meant for the whole world will also have a lot of distortion if you are just viewing and working with the US.
- Choose a projected coordinate system meant for your area of interest.
    - Generally, for maps of the US, choose a conic or conformal projected coordinate system meant for North America
- Explore and Retrieve EPSG codes from
    - https://epsg.io/
    - https://spatialreference.org/
    - EPSG codes (sometimes called a Factory Code in ArcGIS) is a unique 4- to 6-digit code for each coordinate system, datum (and zone, if applicable). It's how CRSs are referenced for layers and in code.

## Adding and Visualizing Data in QGIS

High-level overview of QGIS: https://echo360.org/media/b82666cd-4657-4838-a31e-2ff2ee6b9658/public

### QGIS Hackathon Demo Video: https://youtu.be/UW2KFwxhIPc

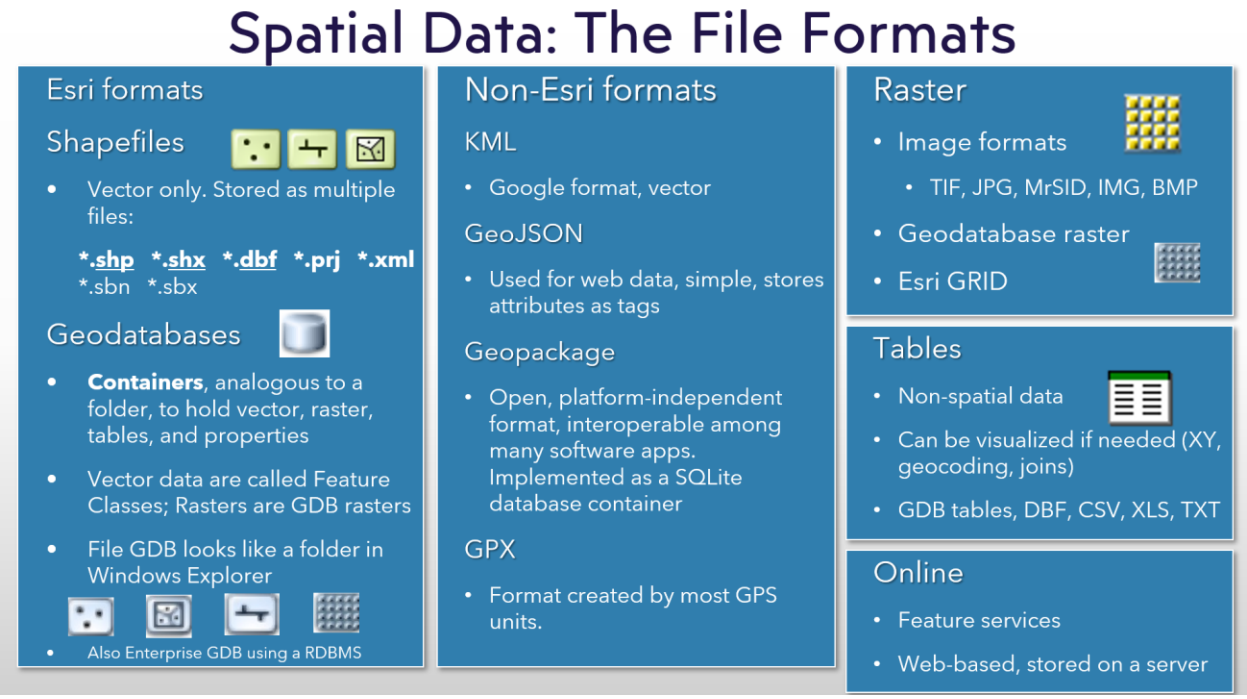
***Covers:***

1. Adding a basemap
2. A quick not on coordinate systems
3. Adding local data
4. Adding online data
5. Symbolizing a layer
6. Adding ArcGIS Online data
7. Where the geoprocessing tools are
8. A quick note on plugins
9. Making a quick map layout

## Common data:

**Typical formats**

- Vector [Points, lines, polygons]: shapefile, geodatabase, geoJSON, KML, Geopackage, online services/feature services
- Raster [cell-based, pixels, array of rows and columns]: TIF, geodatabase raster, IMG, JPG, MrSID and others, online services as imagery or tiled imagery services




## Accessing online data as services:

**Need the REST endpoint / URL**

- Can add to QGIS/ArcGIS Pro or load into a script (ie, with geopandas or rasterio)

***Example from Natural Earth:***

- **Online services**: https://geojson.xyz/
  - Add to QGIS with Add Layer > Vector layer > Protocol > choose HTTP > Paste URL
  - Add into a script using geopandas read_file

```
url = "http://d2ad6b4ur7yvpq.cloudfront.net/naturalearth-3.3.0/ne_110m_land.geojson"
land = geopandas.read_file(url)
land
```

```
url = "http://d2ad6b4ur7yvpq.cloudfront.net/naturalearth-3.3.0/ne_110m_land.geojson"
land = geopandas.read_file(url)
```

    - This will work for any data service with a .geojson URL (Thousands of services available from various websites and portals)
- **Or, download data**: https://www.naturalearthdata.com/downloads/
    - Downloads as a zipped shapefile
    - Point to local data (as .shapefile) and use geopandas read_file

## From ArcGIS Online (AGO)

AGO serves out thousands of layers on many different themes.

Every layer has an Item Details page with details and metadata, like this one for Counties from the USFS:

- https://www.arcgis.com/home/item.html?id=d8d3db28fe72445aa0449cfbcd6d0da3

Every page also provides the REST URL which can be used for adding the features to a map or script.

The Counties REST URL is this:

- https://apps.fs.usda.gov/arcx/rest/services/EDW/EDW_County_01/MapServer/0

In **QGIS**, use the Add Layer > Add ArcGIS REST Service Layer > click New > and provide a name and the URL for the connection.

In **ArcGIS Pro**, use Add Data from Path > paste the service URL

**For a raster / image service, the steps may be slightly different:**

Annual NLCD layer from ArcGIS Online.

Item details:

https://www.arcgis.com/home/item.html?id=32e2ccc6416746a9a72b4d216813f84f

REST URL:

https://di-nlcd.img.arcgis.com/arcgis/rest/services/USA_NLCD_Annual_LandCover/ImageServer

Follow the same steps as the vector ArcGIS REST service (above)

Here's the link from the MRLC page providing direct WMS (web mapping services URLs) for various layers, including Land Cover

https://dmsdata.cr.usgs.gov/geoserver/mrlc_Land-Cover-Native_conus_year_data/wms?

To add this to QGIS, use the Add Layer > WMS option > Make a New connection > enter the URL and click Connect.

Since it's not from an ArcGIS server, it's a different process.